

Given: $v_1 = (1, 1, 1)$, $v_2 = (0, -1, 2)$, $v_3 = (1, 0, 1)$

$T(v_1) = (5, 0, -1)$, $T(v_2) = (-3, 5, -1)$, $T(v_3) = (1, 0, 1)$

1.) Show $(2, -1, 1)$ as a combination of v_1, v_2, v_3 : $(2, -1, 1) = av_1 + bv_2 + cv_3$.

So,
$$\begin{cases} a + c = 2 \\ a - b = -1 \\ a + 2b + c = 1 \end{cases}$$

Then, $c = 2 - a$ $a + 2(a + 1) + (2 - a) = 1$
 $b = a + 1$ $2a + 4 = 1 \rightarrow a = -\frac{3}{2}$

$b = -\frac{3}{2} + 1 = -\frac{1}{2}$, $c = 2 + \frac{3}{2} = \frac{7}{2}$ So, $(2, -1, 1) = -\frac{3}{2}v_1 - \frac{1}{2}v_2 + \frac{7}{2}v_3$

2.) $T(2, -1, 1) = -\frac{3}{2}T(v_1) - \frac{1}{2}T(v_2) + \frac{7}{2}T(v_3)$

$= -\frac{3}{2}(5, 0, -1) - \frac{1}{2}(-3, 5, -1) + \frac{7}{2}(1, 0, 1)$

$= (-\frac{15}{2}, 0, \frac{3}{2}) + (\frac{3}{2}, -\frac{5}{2}, \frac{1}{2}) + (\frac{7}{2}, 0, \frac{7}{2})$

$= (-\frac{5}{2}, -\frac{5}{2}, \frac{11}{2})$